Feature Visualization

Neural Networks understanding of Images

Prof. Anil Singh Parihar

Department of Computer Science & Engineering

Delhi Technological University, Delhi

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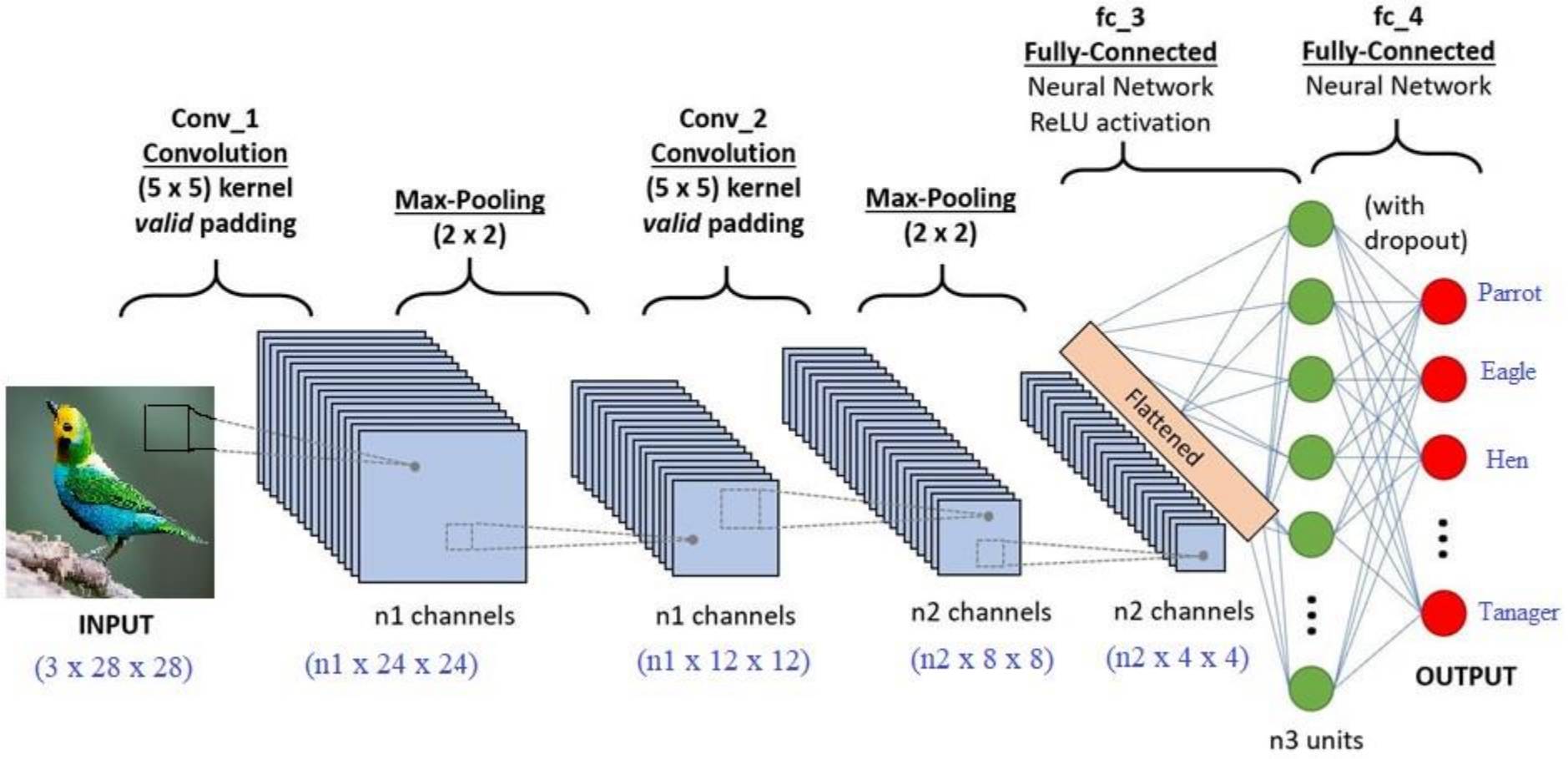
Preliminaries: Typical CNN classifier



Convolution

These are the network parameters to be learned.

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

⋮ ⋮

Each filter detects a small pattern (3 x 3).

Convolution

Stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

Dot product → 3

Filter 1

1	-1	-1
-1	1	-1
-1	-1	1

Convolution

Stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

Dot product →

3 -1

Filter 1

1	-1	-1
-1	1	-1
-1	-1	1

Convolution

If stride=2

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

3

Filter 1

1	-1	-1
-1	1	-1
-1	-1	1

Convolution

If stride=2

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

3 -3

Filter 1

1	-1	-1
-1	1	-1
-1	-1	1

Convolution

Stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

Filter 1

1	-1	-1
-1	1	-1
-1	-1	1

Convolution

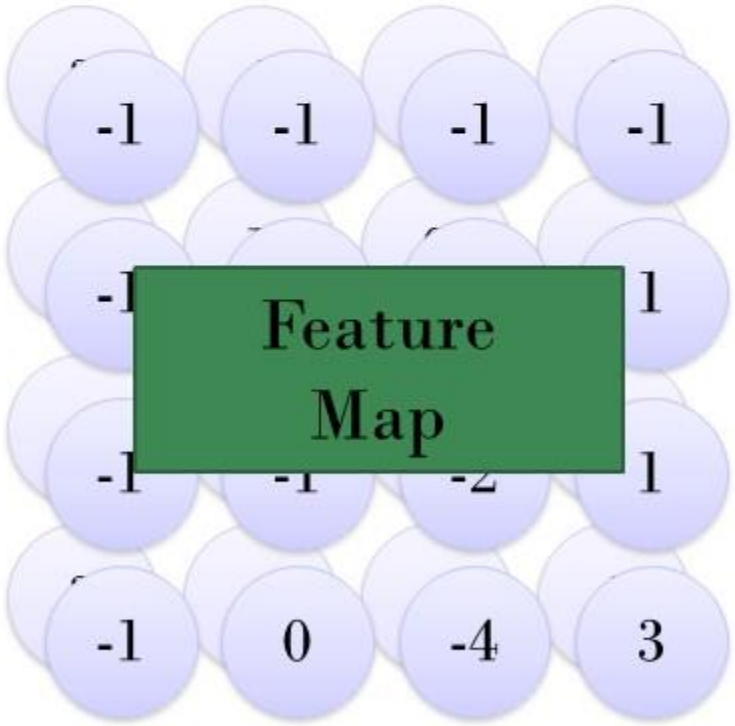
Two 4 x 4 feature maps
Forming 2 x 4 x 4 Tensor

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

Repeat this for each filter

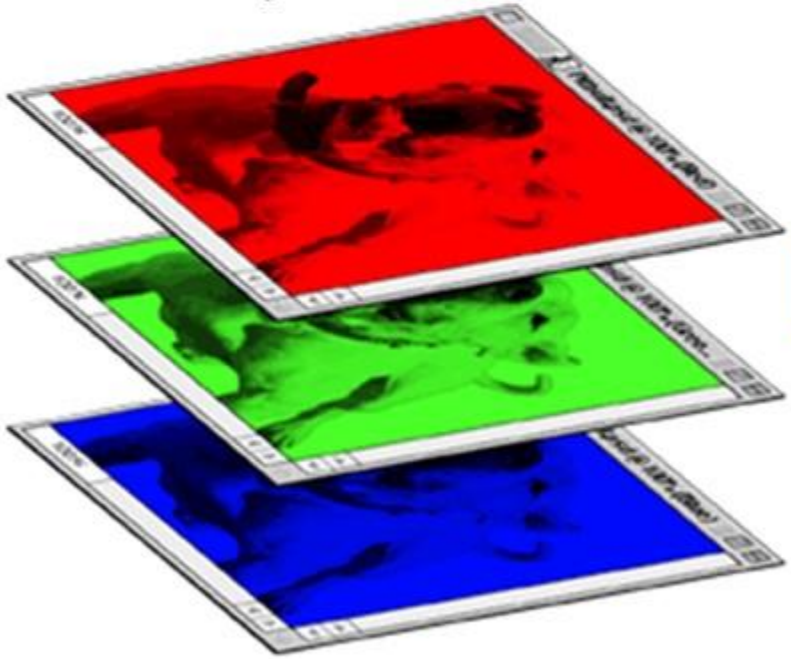


Filter 2

-1	1	-1
-1	1	-1
-1	1	-1

Color image: RGB 3 channels

Color image



1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

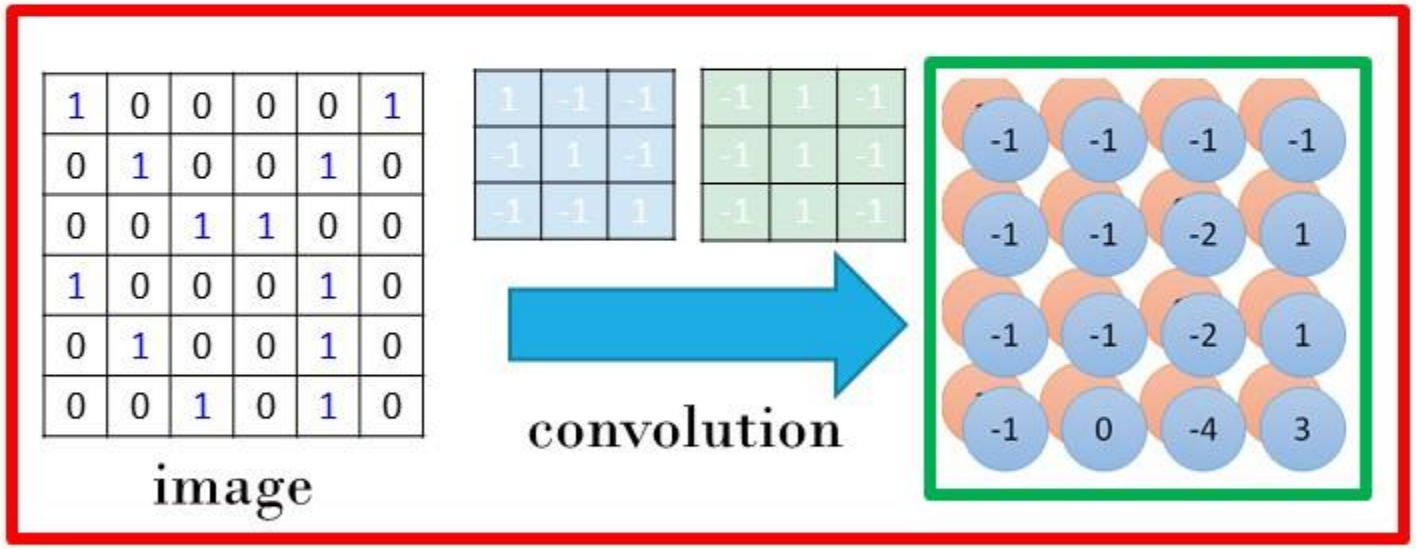
1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

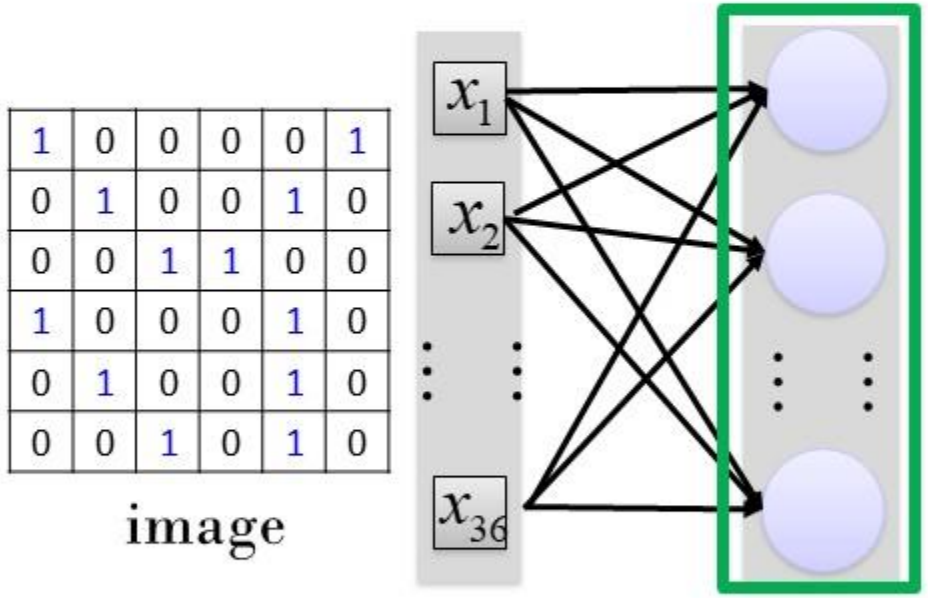
-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

Convolution v.s. Fully Connected



Convolution



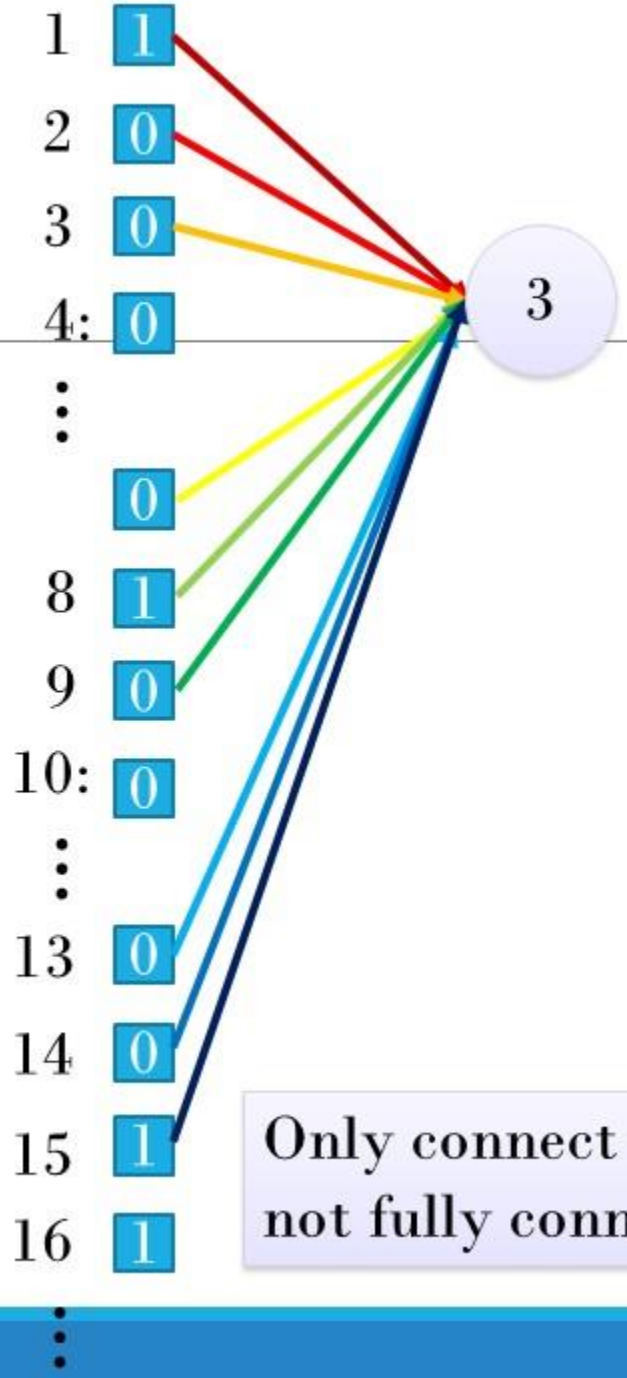
Fully-connected

Filter 1

1	0	0	0	0	
0	1	0	0	1	
0	0	1	1	0	
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

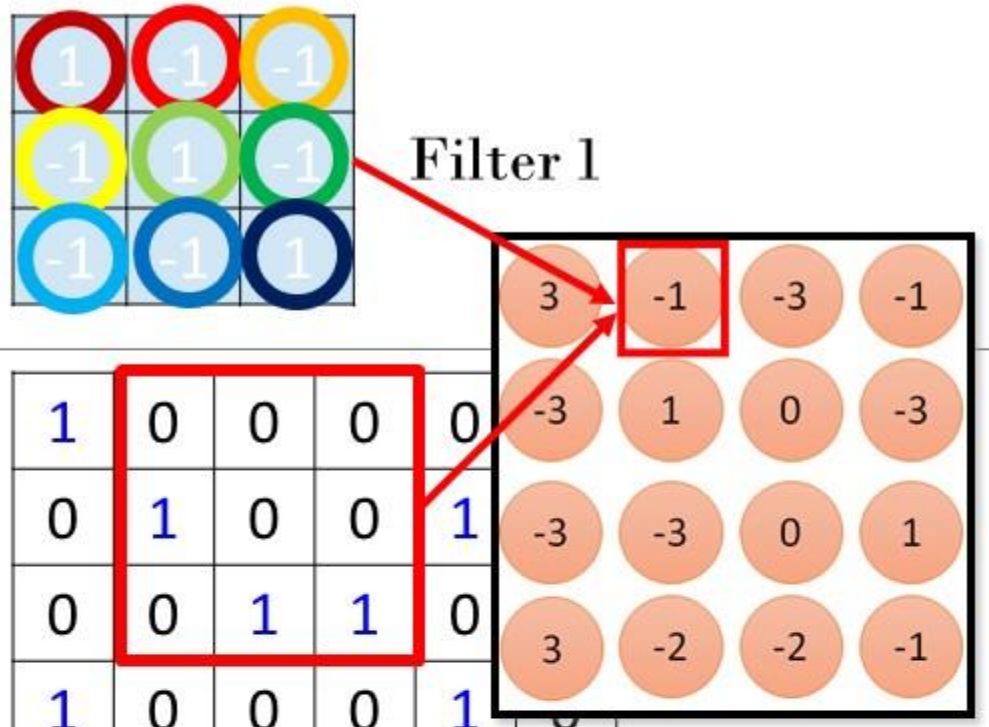
6 x 6 image

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

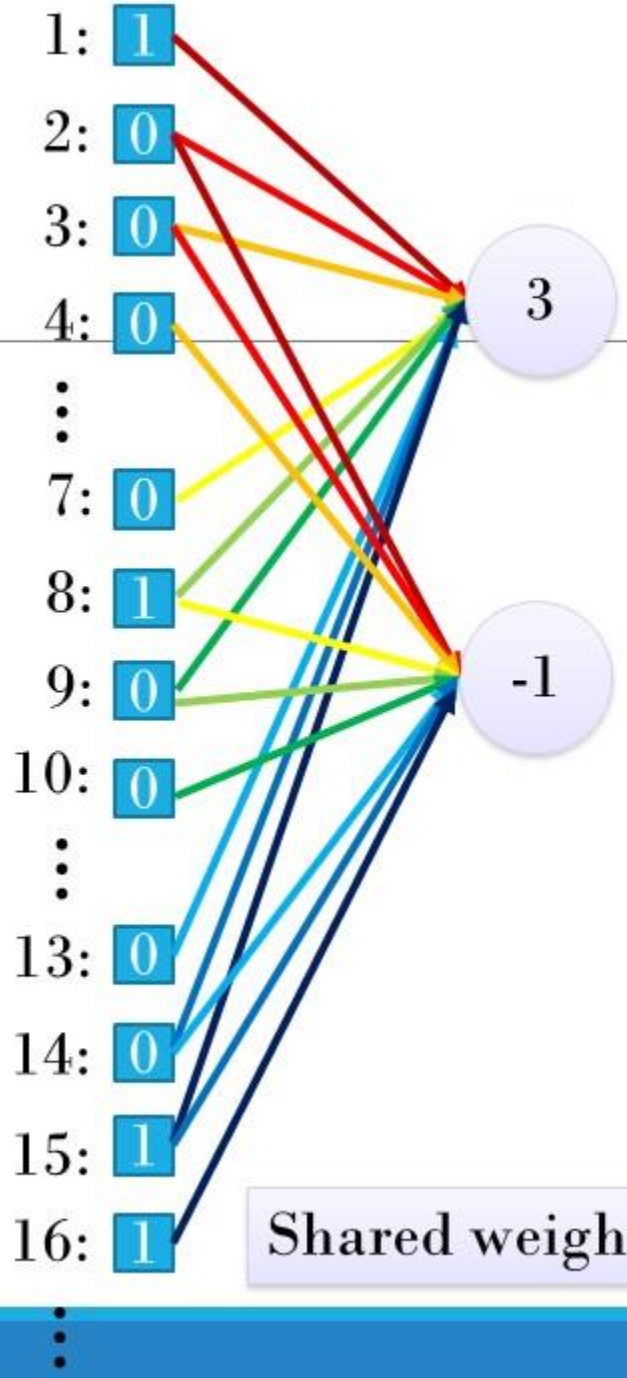


fewer parameters!

Only connect to 9 inputs, not fully connected



6 x 6 image



Fewer parameters

Even fewer parameters

Shared weights

Pooling

Subsampling pixels will not change the object

bird

bird

We can subsample the pixels to make image smaller

→ fewer features to characterize the image

Max Pooling

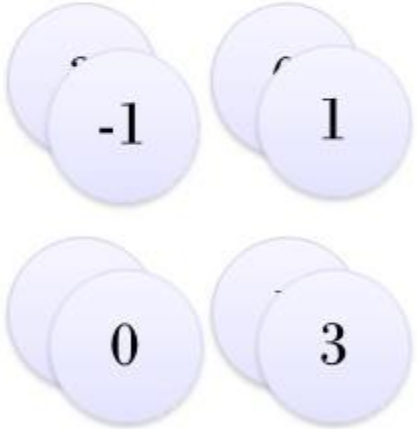
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

Conv

Max Pooling

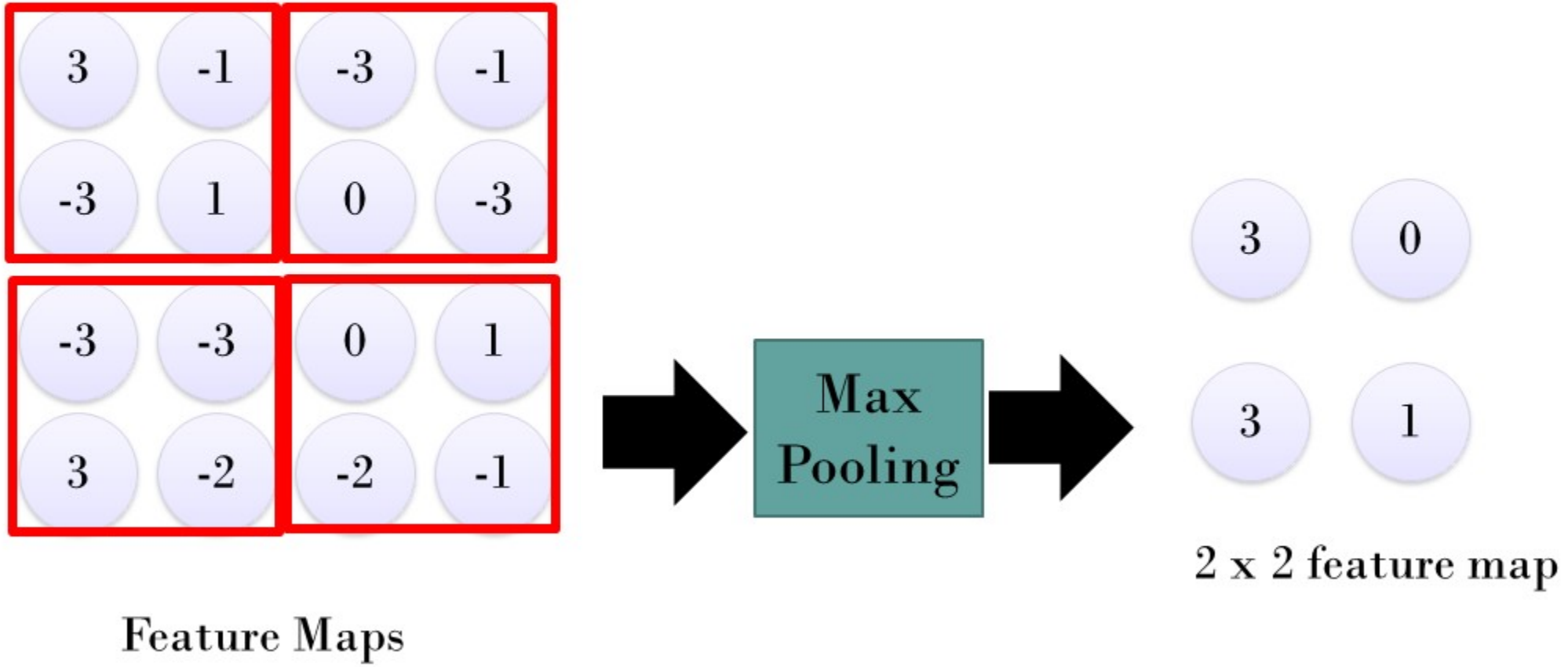
New feature maps but smaller



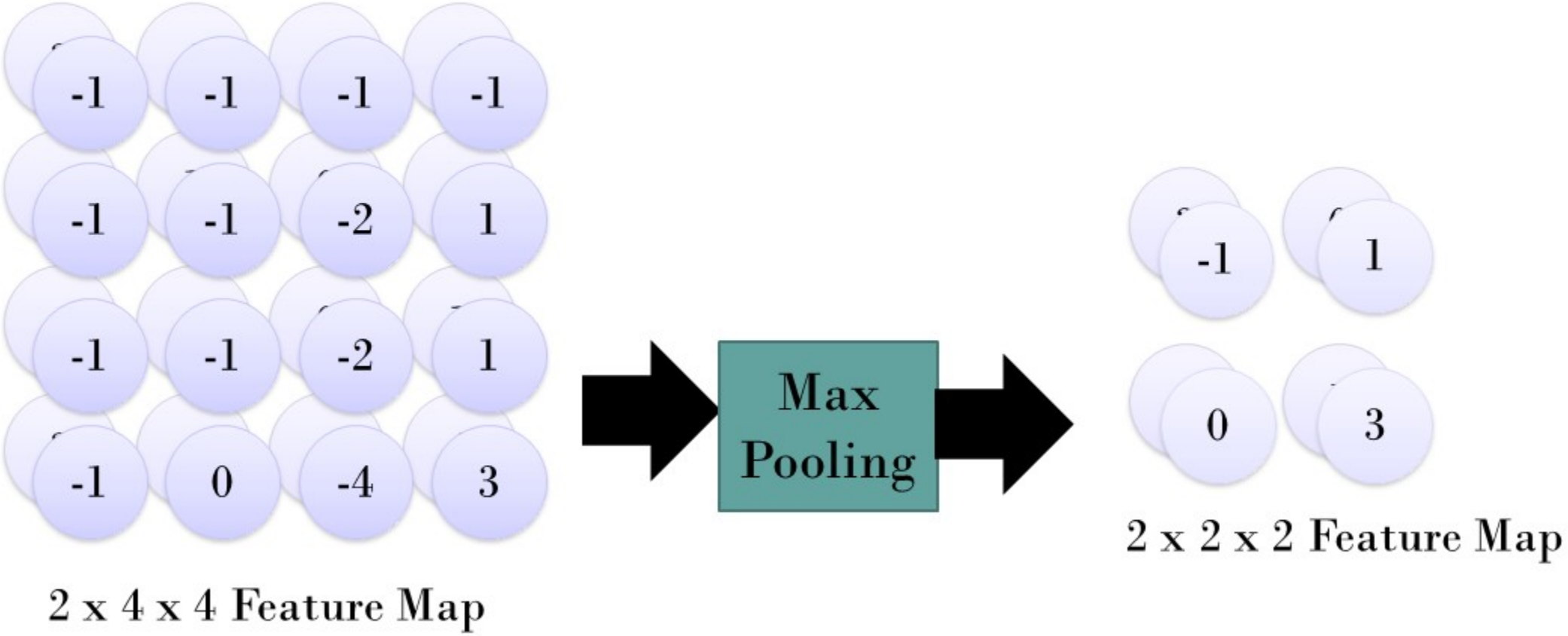
2 x 2 image

Each filter is a channel

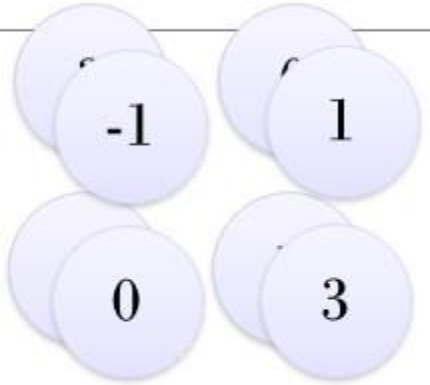
Max Pooling


3
-1
-3
-1
-3
1
0
-3
-3
-3
0
1
3
-2
-2
-1
Max Pooling
3
0
3
1
2 x 2 feature map
Feature Maps

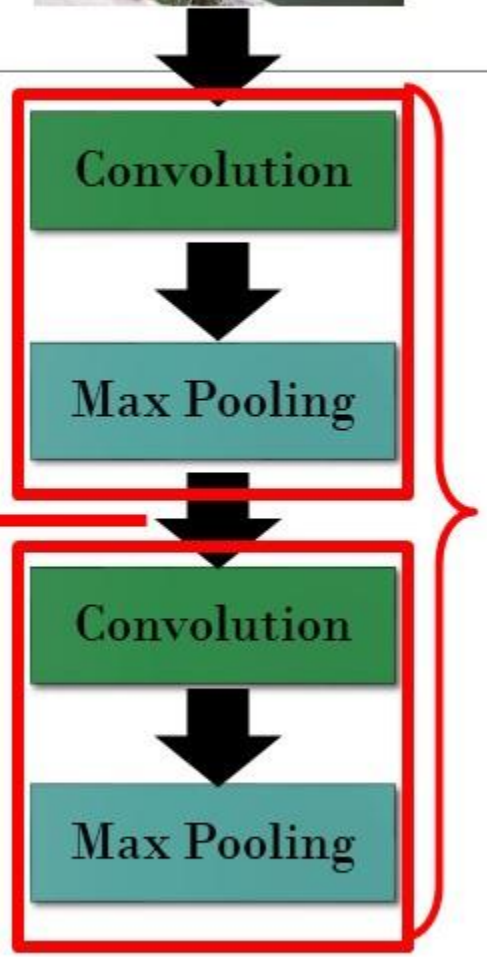
Max Pooling


-1 -1 -1 -1
-1 -1 -2 1
-1 -1 -2 1
-1 0 -4 3
Max Pooling
-1 1
0 3
2 x 2 x 2 Feature Map
2 x 4 x 4 Feature Map

Feature map from Layer 1



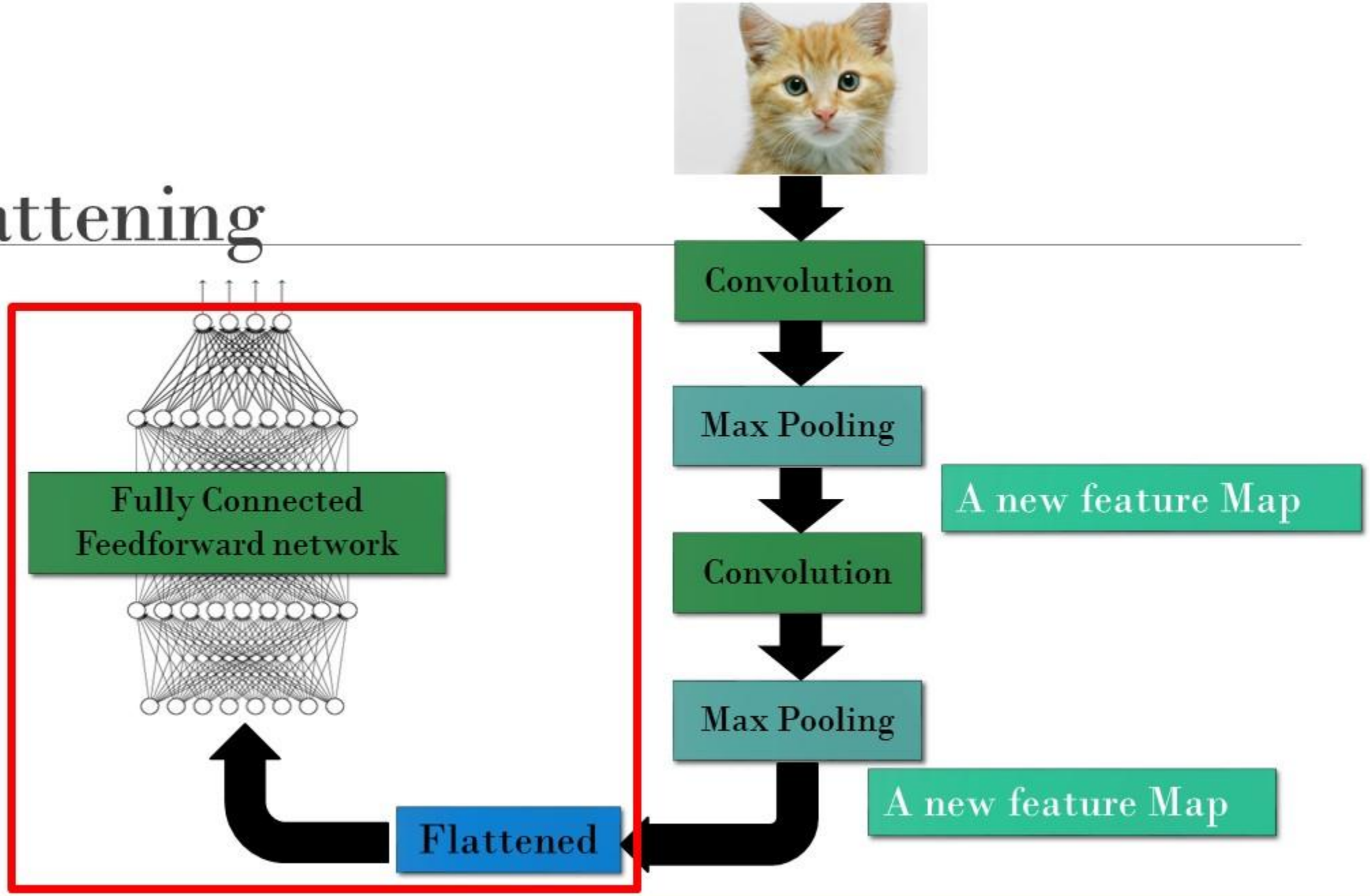
A new feature map



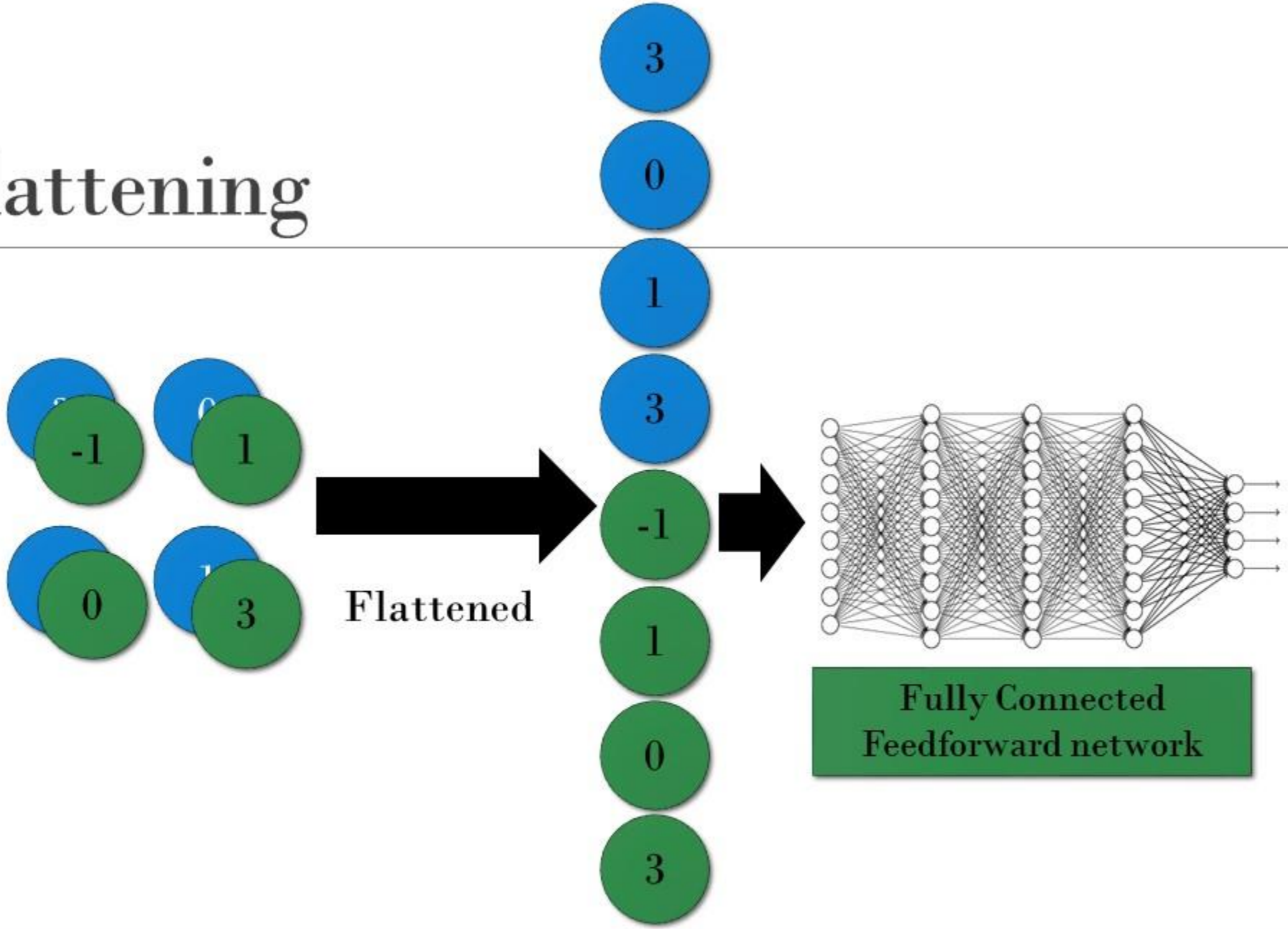
Can repeat many times

- Smaller than the original image
- The number of channels is the number of filters

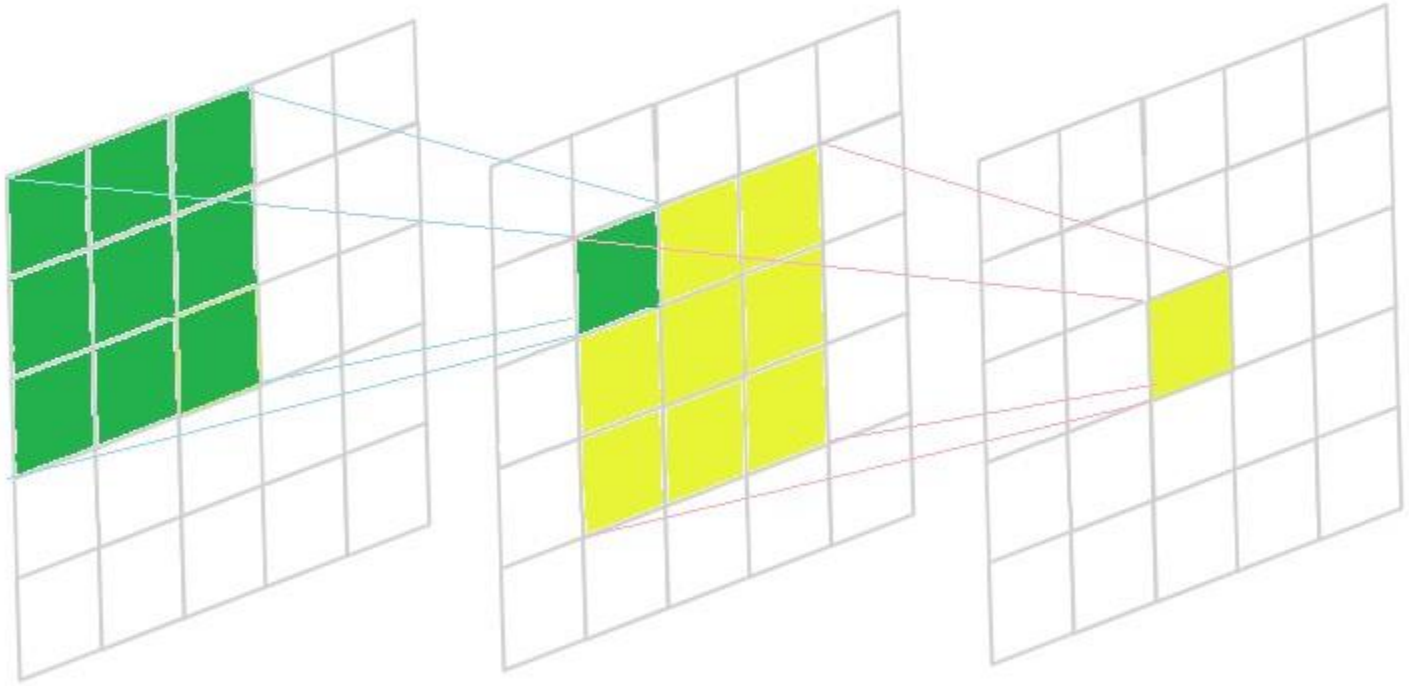
Flattening


Convolution
Max Pooling
A new feature Map
Convolution
Max Pooling
A new feature Map
Fully Connected
Feedforward network
Flattened

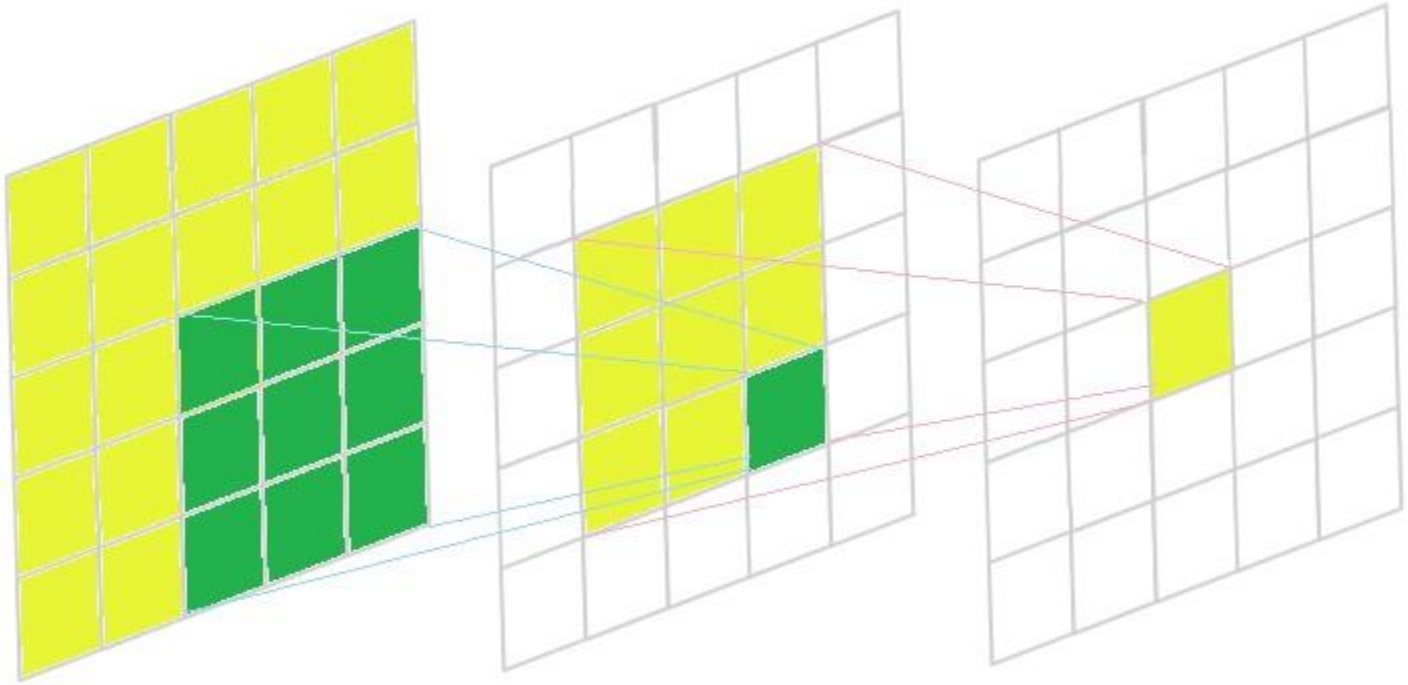
Flattening


-1
1
0
3
Flattened
3
0
1
3
-1
1
0
3
Fully Connected
Feedforward network

Receptive Field



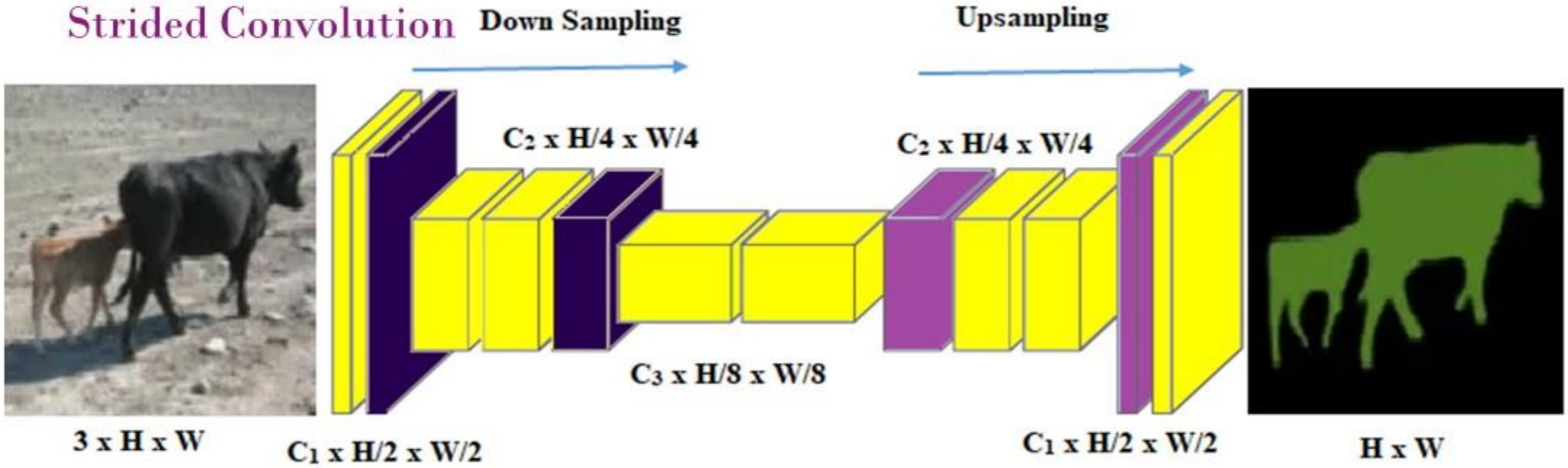
Receptive Field



A CNN model for semantic segmentation

Downsampling:
Pooling,
Strided Convolution

Upsampling: ??



Upsampling

Popular Approaches:

- Unpooling: Nearest neighbour interpolation, bilinear interpolation
- Max Unpooling (uses position of max element during max pooling)
- Transpose Convolution (learnable upsampling)

Unpooling

Nearest Neighbor

3	4
9	7

Input: 2 x 2

3	3	4	4
3	3	4	4
9	9	7	7
9	9	7	7

Output: 4 x 4

Unpooling

Bed of Nails

3	4
9	7

Input: 2 x 2

3	0	4	0
0	0	0	0
9	0	7	0
0	0	0	0

Output: 4 x 4

Max Unpooling

Bed of Nails

2	8	1	2
0	4	4	9
6	1	7	1
5	3	3	0

Input: 4 x 4

8	9
6	7

Output: 2 x 2

Network blocks

4	7
9	5

Input: 2 x 2

0	4	0	0
0	0	0	7
9	0	5	0
0	0	0	0

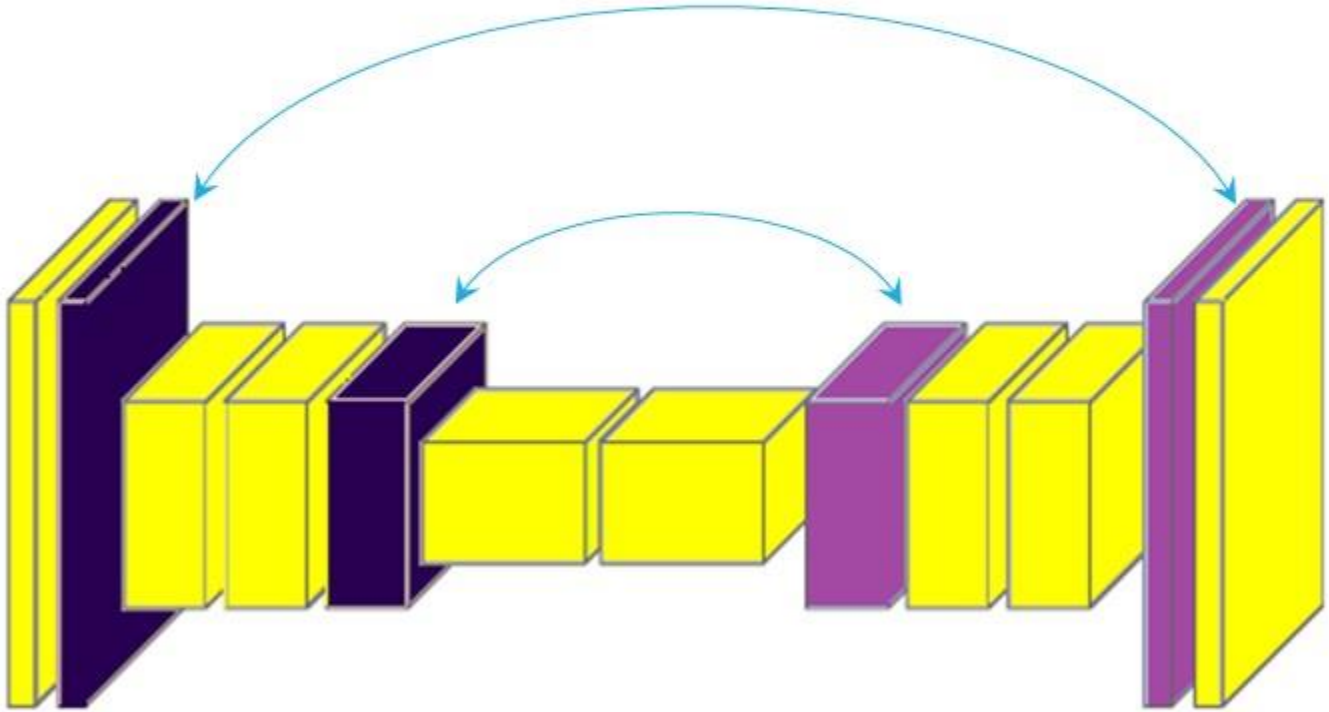
Output: 4 x 4

Max Pooling:
Remembers positions of max elements using a switch

Max Unpooling:
Uses positions of max elements (switch) from corresponding pooling layer

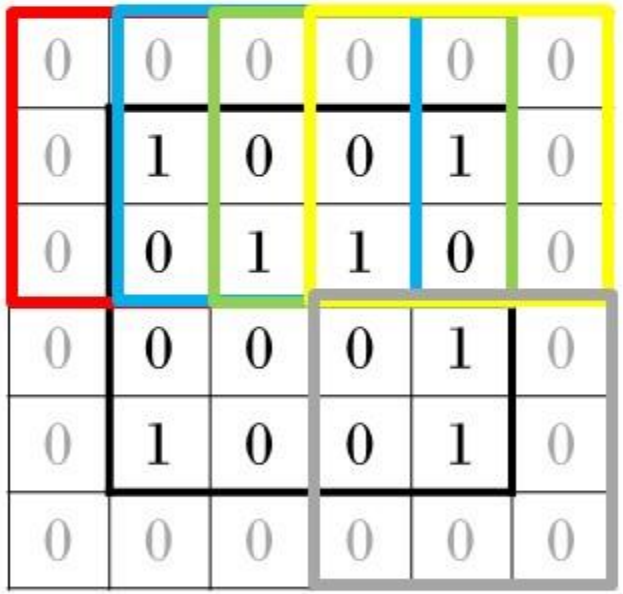
Max Unpooling

Pairs Downsampling & Upsampling Layers



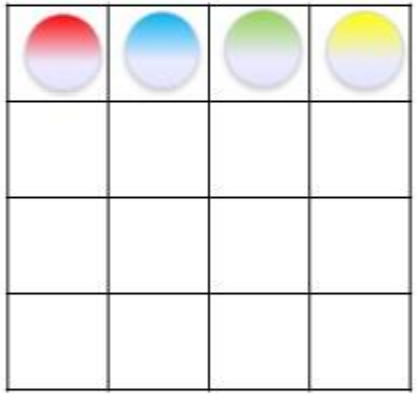
Transpose Convolution

Recall normal convolution with stride = 1, pad =1 & 3x3 filter



Input: 4 x 4

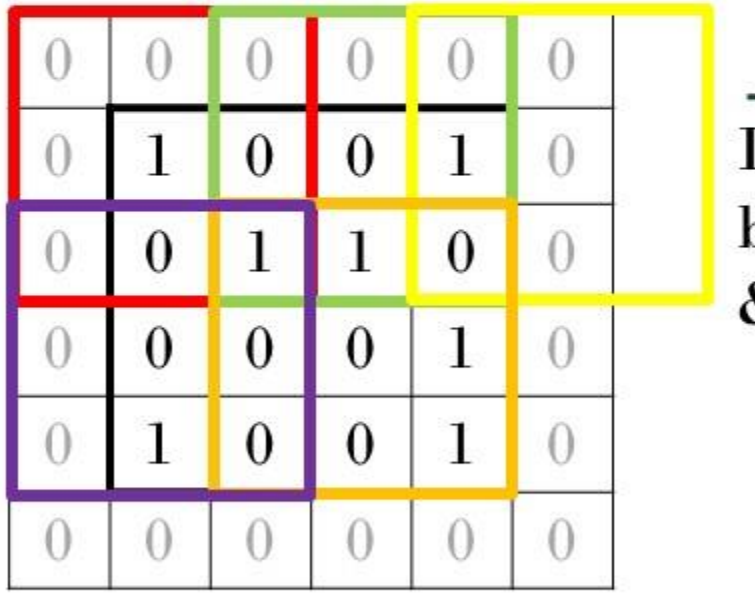
Dot product between input & filter



Output: 4 x 4

Transpose Convolution

Recall normal convolution with stride = 2, pad =1 & 3x3 filter



Input: 4 x 4

Dot product between input & filter

Output: 2 x 2

Note: Stride gives ratio of input to output

Transpose Convolution

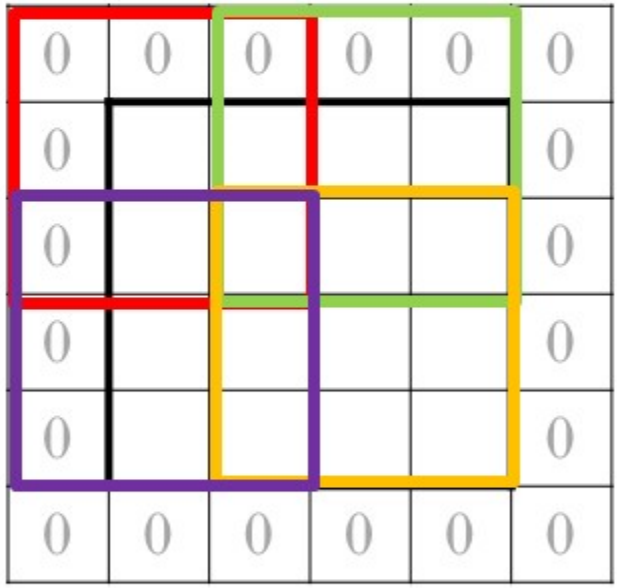
Also Called:

- Deconvolution (misnamor)
- Upconvolution
- Fractionally strided convolution

Transpose convolution with stride = 2, pad =1 & 3x3 filter

Input: 2 x 2

Input is multiplied as a weight to filter elements (learnable)



Output: 4 x 4

Note: Stride gives ratio of Output to Input

Transpose Convolution performs upsampling with learnable parameters

For more details: Vincent Dumoulin, Francesco Visin "A guide to convolution arithmetic for deep learning" 2018